Selection of SDLC Model

Course: Information System Design & Software Engineering Lab
Course Code: CSE 346

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Goals-

- ✓ Understand the concept of Software Development Life Cycle (SDLC)
- ✓ Identify different SDLC models
- ✓ Analyze project requirements
- ✓ Select the most suitable SDLC model for their project

Concept 1: **What is SDLC?**

Software Development Life Cycle (SDLC) is a structured process used to design, develop, test, and maintain software systems.

It provides a **systematic approach** to software development.

Main Phases of SDLC

1. Planning
2. Requirement Analysis
3. System Design
4. Implementation
5. Testing
6. Deployment
7. Maintenance

Why Do We Need SDLC?

Without a structured development process, software projects may face:

- Poor planning
- Unclear requirements
- Budget overruns
- Delayed delivery
- Low-quality software

Benefits of SDLC

- Better project management
- Improved software quality
- Reduced development risk
- Clear development stages

Concept 2: What is an SDLC Model?

An SDLC Model is a framework that defines **how the development phases are organized and executed**.

Different projects require **different development approaches**.

Examples of SDLC Models

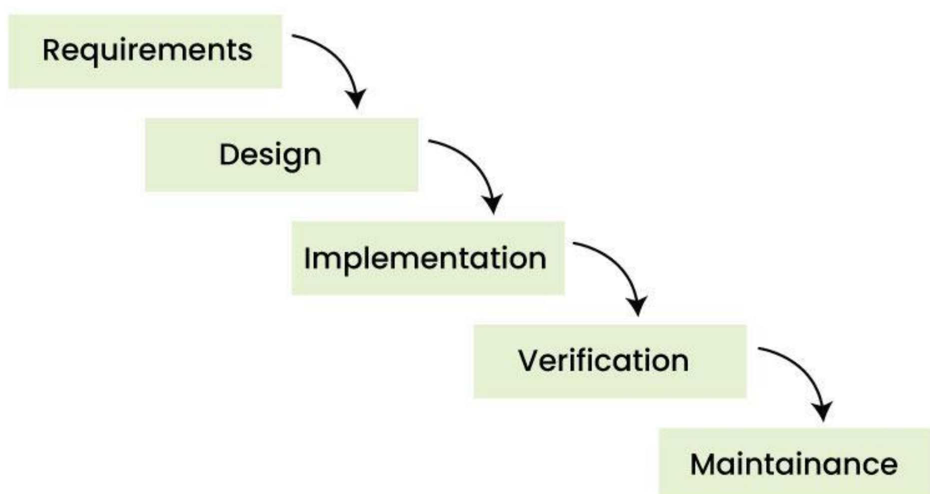
- Waterfall Model
- Iterative Model
- Incremental Model
- Spiral Model
- Agile Model
- V-Model

Model 1: Waterfall Model

The **Waterfall Model** is a **linear and sequential approach** where each phase must be completed before moving to the next phase.

Phases

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Deployment
6. Maintenance



Waterfall Model

Advantages

- Easy to understand
- Well structured
- Suitable for small projects

Disadvantages

- Difficult to change requirements later

When to Use the Waterfall Model

Use this model when:

- **Project requirements are clearly defined**
- Requirements **will not change frequently**
- The system is **simple or moderately complex**
- The project has **a fixed timeline and budget**
- The development team prefers a **structured process**

Example Scenario

University Library Management System

Reason:

- Requirements are clear
- Features are well known
- System complexity is low

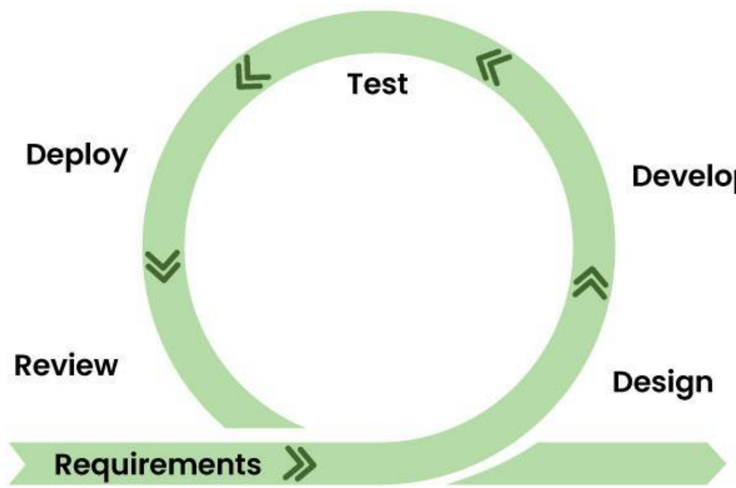
Model 2: Agile Model

The **Agile Model** focuses on **flexibility and iterative development**.

Development happens in **small cycles called sprints**.

Characteristics

- Continuous improvement
- Frequent user feedback
- Faster delivery



Agile Model

Advantages

- Flexible development
- Faster response to change

Disadvantages

- Requires experienced team
- Harder to manage large projects

When to Use the Agile Model

Use Agile when:

- Requirements **change frequently**
- Customer feedback is important
- Fast development and **frequent updates** are required
- The project is **user-focused**
- The team can work in **short development cycles**

Example Scenario

Online Food Ordering System

Reason:

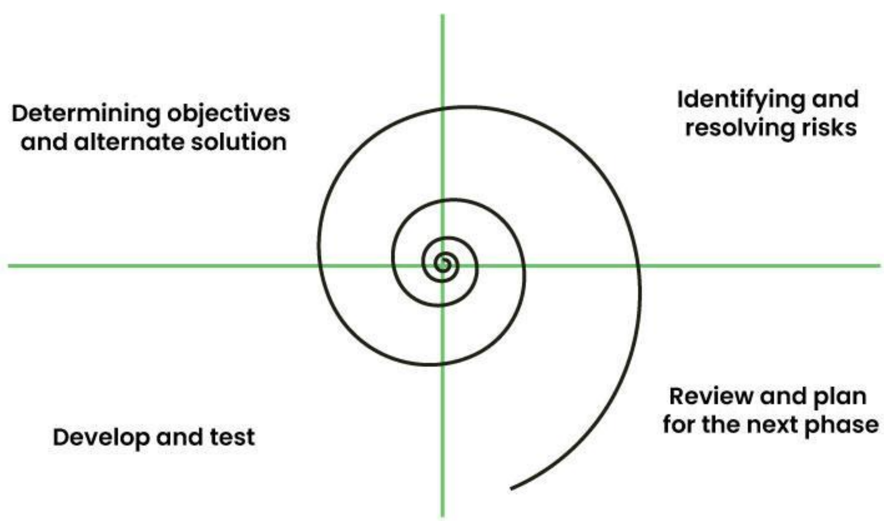
- Features may change
- Customer feedback is important
- New updates may be added frequently

Model 3: Spiral Model

The **Spiral Model** combines **iterative development with risk analysis**. The system is developed in **multiple cycles**, and risks are analyzed in every stage.

Main stages:

- Planning
- Risk Analysis
- Development
- Evaluation



Spiral model

When to Use the Spiral Model

Use this model when:

- The project is **large and complex**
- There are **high risks involved**
- Requirements may evolve over time
- The system needs **continuous risk evaluation**

Example Scenario- Banking System

Reason:

- High security requirements
- Complex system features
- Risk analysis is critical

Model 4: V-Model

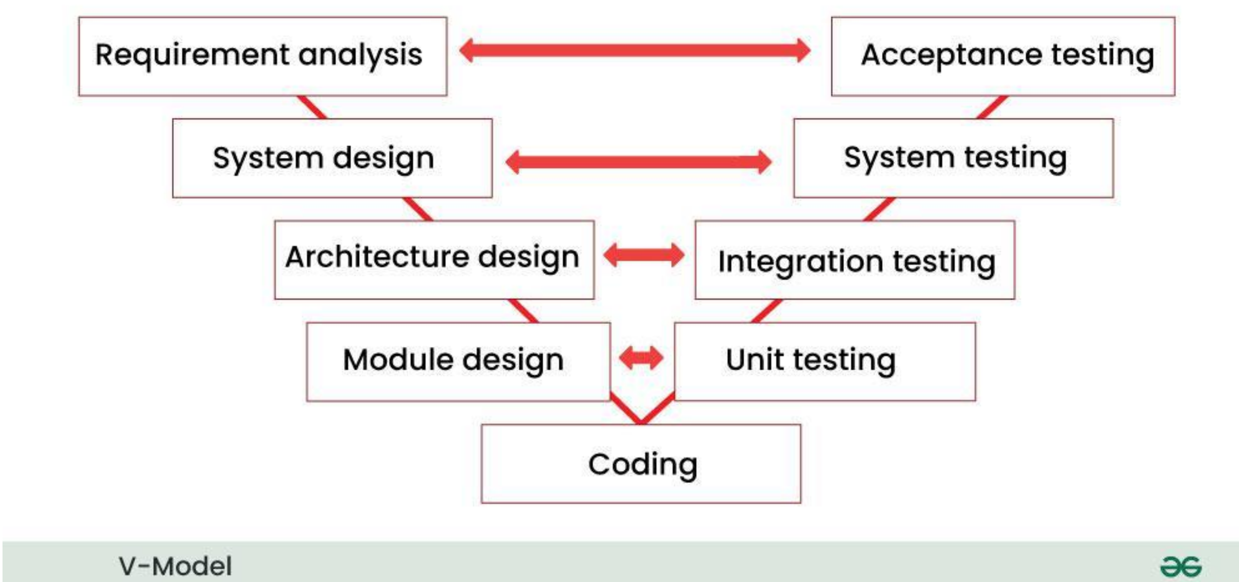
Description

The **V-Model (Verification and Validation Model)** emphasizes **testing at every stage of development**.

Each development phase has a **corresponding testing phase**.

Example:

- Design → Unit Testing
- System Design → Integration Testing
- Requirement Analysis → Acceptance Testing



When to Use the V-Model

Use this model when:

- Requirements are **very clear and stable**
- The system requires **high reliability**
- **Testing and validation are critical**
- The project involves **strict quality standards**

Example Scenario

Hospital Management System

Reason:

- System accuracy is important
- Data reliability is critical
- Errors must be minimized

Quick comparison

Model	Best For	Example
Waterfall	Clear requirements	Library System
Agile	Changing requirements	Mobile apps
Spiral	High-risk systems	Banking
V-Model	High reliability	Healthcare

To-do:

1. Project Name
2. Project Description (Short)
3. Selected SDLC Model
4. Justification